

CrossPiezo: Two High-Throughput Databases Agree on Broad Promising Regions but Cannot Resolve the Elite Tail in Piezoelectric Screening

Anonymous Author(s)¹

¹Anonymous Institution

2 August 2026

Abstract

High-throughput piezoelectric databases are widely used to train and benchmark machine-learning models, yet their agreement is usually reported as a single global correlation. Here we reframe comparison between two independently generated high-throughput datasets as a question of *screening resolution*: do the Materials Project (MP) and JARVIS databases agree on broad promising regions of chemical space, yet fail to reproducibly resolve the elite tail of candidates? Using the frozen CrossPiezo-Invariant-v1 benchmark of 573 strict structure-matched pairs (P0) and 207 tighter matches (P2), we compute coordinate-invariant response scalars F1, F3 and F4 and sweep the screened quantile from 1% to 50%. The full-range normalised area under the chance-adjusted concordance curve (nAUCC) for F1 on P0 is 0.105, and the simultaneous 95% lower confidence bound of adjusted overlap first exceeds 0 at $q = 33\%$; no five-quantile persistent onset is observed at $\delta = 0.05$, showing that chance-adjusted agreement is not reliably sustained in the elite tail. We report partial nAUCC separately for the elite (1–10%), intermediate (10–20%) and broad (20–50%) bands; concordance is weakest in the elite band and only emerges after substantially broadening the candidate pool. Corrected high-response sensitivity analyses avoid the pooled-selection collider bias: dual-high, JARVIS-anchor and MP-anchor subsets all show reduced or negative elite-tail concordance, confirming that the gap is not an artifact of the selection definition. Simpler scalar controls (unit-cell volume, band gap) are markedly more cross-source consistent than the piezoelectric response, with positive $\Delta\tau$ and ΔnAUCC ; their higher consistency reflects lower workflow sensitivity in these two sources, not proof that response disagreement is mainly conventional. Equal-budget source-robust portfolios improve worst-source recall and NDCG relative to JARVIS-only or MP-only selection, and paired-bootstrap confidence intervals on the difference are reported. The balanced-union strategy reaches worst-source recall of 1.0 at twice the elite budget by coverage construction; this is an upper bound, not a method gain. All numbers are hash-bound and traceable to the Phase 7C manifest and the Phase 8A audit; a 48-material third-protocol adjudication plan is pre-registered for independent validation.

1 Introduction

Piezoelectric response tensors are central to electromechanical materials design, yet costly to compute with density-functional perturbation theory (DFPT) [1, 2]. The Materials Project (MP) and JARVIS projects have released thousands of computed tensors [3, 4], and recent equivariant neural networks report strong in-source prediction accuracy [5, 6, 7]. Improved agreement with a held-out subset of

one database does not guarantee that a candidate is robustly ranked across independently generated high-throughput datasets [8, 9, 10].

We introduce *screening resolution* as a diagnostic lens. Rather than asking “what is the average rank correlation between two sources?” we ask “at what elite fraction q does the overlap between source-specific top- q sets become statistically distinguishable from chance?” This distinction matters for materials discovery: a model that is validated only on aggregate correlation may still disagree with an independent source on the very candidates a human experimenter would pursue [11, 12, 13].

2 Screening-resolution framing

Paired universe. We use the frozen Phase 6A panel: P0 ($n = 573$) and P2 ($n = 207$), structure-matched pairs with identical or near-identical crystal structures across MP and JARVIS [14, 15]. We do not modify panel membership. The panels were built by structure matching of relaxed CIFs followed by manual inspection of the top candidates; full details and the frozen membership file are released in the repository. At the 1% elite quantile the effective sample is approximately 6 materials in P0 and 2 materials in P2, so elite-tail statements should be read as lower-bound diagnostics rather than precise population estimates.

Response functionals. For each source we compute three coordinate-invariant scalars: F1 = Cartesian Frobenius norm $\|\mathbf{e}\|_F$; F3 = true collinear longitudinal maximum $\max_{\|\mathbf{n}\|=1} |\mathbf{n}_i e_{ijk} \mathbf{n}_j \mathbf{n}_k|$ [16]; F4 = Kelvin/Mandel operator norm on symmetric strain. F3 is reported because it isolates the longitudinal single-crystal response; F1 and F4 are more relevant for polycrystalline or norm-based screening.

Chance-adjusted concordance curve. For each panel, metric and quantile $q = 1\%, \dots, 50\%$, we report the observed top- q Jaccard overlap J_q , the exact hypergeometric null expectation $\mathbb{E}[J_q]$ under independent rankings, and the chance-adjusted overlap

$$\tilde{J}_q = \frac{J_q - \mathbb{E}[J_q]}{1 - \mathbb{E}[J_q]}. \quad (1)$$

We construct a simultaneous 95% confidence band around $\tilde{J}_q$ using a paired bootstrap with a studentized sup-norm critical value [17, 18]. The normalised area under the concordance curve is

$$\text{nAUCC} = \frac{1}{q_{\max} - q_{\min}} \int_{q_{\min}}^{q_{\max}} \tilde{J}_q dq, \quad (2)$$

and the persistent consensus onset is the smallest q for which the lower band exceeds a threshold δ for at least five consecutive quantiles. nAUCC is a descriptive summary of the curve; uncertainty is conveyed by the simultaneous bands on the underlying curve, not by a confidence interval on the area itself.

Banded reporting. To avoid unified claims about broad agreement, we split the 1–50% curve into elite (1–10%), intermediate (10–20%) and broad (20–50%) bands and report partial nAUCC and simultaneous bands for each.

Corrected high-response sensitivity. Phase 7B pooled the two sources to define a high-response subset, which can induce collider bias. Phase 7C uses three corrected definitions: (i) `dual_high`: both source values exceed the $1 - f$ quantile of $\min(F_J, F_M)$; (ii) `jarvis_anchor`: top f by JARVIS, evaluated on MP; (iii) `mp_anchor`: top f by MP, evaluated on JARVIS. The legacy pooled selection is retained only as an exploratory check with a conditional permutation that re-executes the selection after shuffling source labels [17]. We report $f = 0.10$ and $f = 0.25$.

Controls and provenance. We compare piezoelectric response against volume, band gap, energy above hull and dielectric trace. For each attribute we record source field, source database, unit, calculation type/version, missingness and a flag for possible same-field copying (more than 5% identical JARVIS/MP values) [19]. We report $\Delta\tau = \tau_{\text{control}} - \tau_{\text{piezo}}$ and ΔnAUCC with grouped paired-bootstrap confidence intervals, permutation p -values and Benjamini–Hochberg FDR [20]. Controls show that response properties have higher workflow sensitivity in these two sources; they do not prove that disagreement is mainly conventional.

Robust portfolio benchmark. We fix an elite fraction $q^* = 10\%$. Primary results use an equal budget $b = 1.0$; $b = 1.5$ and $b = 2.0$ are reported as budget sensitivity. Strategies include single-source, average percentile, Borda count, maximin percentile, intersection-first, balanced union, disagreement abstention and a greedy minimax oracle [21, 22]. All deterministic strategies are evaluated on the full panel; disagreement abstention tunes its penalty with nested grouped cross-validation. We also perform 5-fold grouped cross-evaluation. Improvements over the better of JARVIS-only and MP-only are reported as paired differences with grouped paired-bootstrap confidence intervals, not only point estimates. Full-panel differences are modest and their confidence intervals include zero; they should be read as decision examples rather than as proof of large method gains. The balanced-union strategy at $b = 2.0$ is labelled a *coverage upper bound* because it contains the union of the two source top- q^* sets by construction.

3 Results

3.1 Screening-resolution curves

Figure 1 shows the screening-resolution curves for P0. Table 1 summarises the corresponding concordance diagnostics. On P0 the three functionals behave similarly: full-range nAUCC is 0.105 for F1, 0.110 for F3 and 0.106 for F4. The pointwise lower confidence bound first exceeds 0 at 33% for F1, but a five-quantile persistent onset at $\delta = 0.05$ is not observed for F1, and the P2 panel shows even weaker concordance (F1 nAUCC = 0.060). Thus, even if one screens half of the universe, the two sources only barely agree more than would be expected by chance.

Figure 2 and table 2 report partial nAUCC by band. Concordance is weakest in the elite band, modest in the intermediate band, and only reaches clearly positive values in the broad band for F3/F4. This pattern shows that concordance emerges only after substantially broadening the candidate pool.

3.2 Corrected high-response sensitivity

Table 3 shows the corrected high-response analyses on P0 F1. The dual-high subset ($f = 0.10$, $N = 58$) has nAUCC = -0.046, demonstrating that the elite-tail resolution gap is not a by-product of the old pooled-selection definition. The anchor analyses and the exploratory pooled conditional

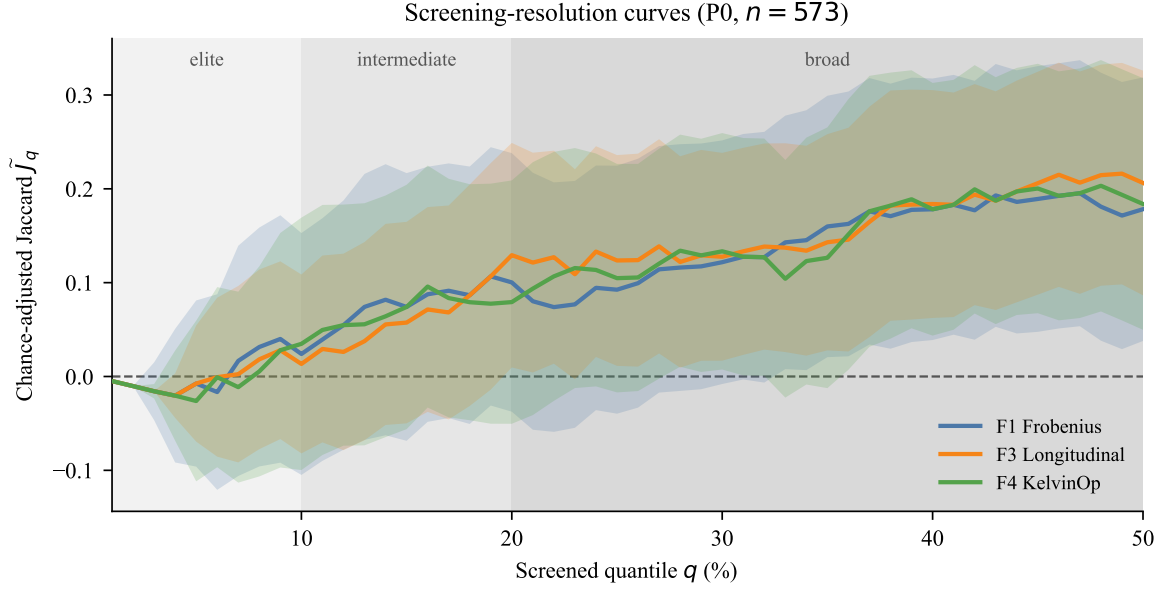


Figure 1: Screening-resolution curves for P0 ($n = 573$). Solid lines show the chance-adjusted Jaccard overlap $\tilde{J}_q$ for F1 (blue), F3 (orange) and F4 (green); shaded ribbons are simultaneous 95% confidence bands. The horizontal dashed line marks chance-adjusted zero overlap. Background shading distinguishes the elite (1–10%), intermediate (10–20%) and broad (20–50%) bands. Concordance is weak in the elite tail and only becomes positive after substantially broadening the candidate pool.

Table 1: Screening-resolution summary for P0 ($n = 573$) and P2 ($n = 207$). $q_{0.00}$ is the pointwise first crossing; $q_{0.05}^{\text{persist}}$ requires five consecutive lower-confidence bounds above 0.05.

Panel	Metric	nAUCC	$q_{0.00}$	$q_{0.05}^{\text{persist}}$
P0	F1_Frobenius	0.105	33%	–
P0	F3_Longitudinal	0.110	20%	38%
P0	F4_KelvinOp	0.106	28%	42%
P2	F1_Frobenius	0.060	–	–
P2	F3_Longitudinal	0.043	–	–
P2	F4_KelvinOp	0.046	–	–

Table 2: Banded partial nAUCC for P0 F1. Elite = 1–10%, intermediate = 10–20%, broad = 20–50%.

Band	Range	Partial nAUCC
elite	1–10%	0.003
intermediate	10–20%	0.076
broad	20–50%	0.145

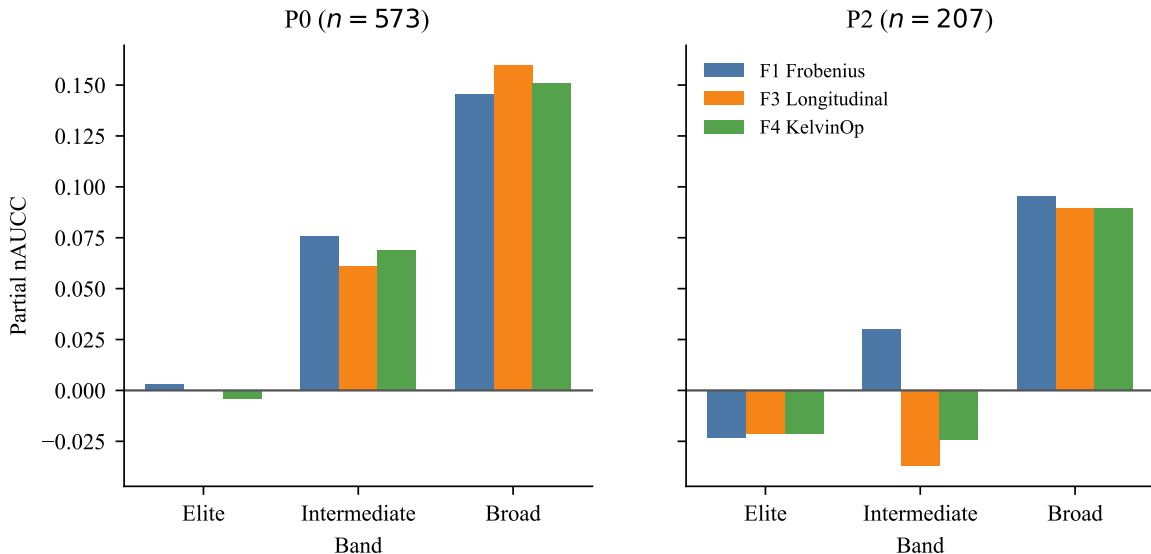


Figure 2: Partial nAUC by band for P0 ($n = 573$) and P2 ($n = 207$). Positive values indicate agreement better than chance. For all metrics, concordance is weakest in the elite band and grows only after broadening the candidate pool.

permutation give consistent results. Because the pooled selection can suffer collider bias, its negative nAUC is reported as exploratory and is not used as standalone proof of a physical mechanism.

Table 3: Corrected high-response sensitivity for P0 F1 at $f = 0.10$.

Method	N	nAUC
dual_high	58	-0.046
jarvis_anchor	57	0.053
mp_anchor	57	-0.018
pooled_exploratory	58	-0.155

3.3 Property controls

Figure 3 and table 5 show that all available controls are more cross-source consistent than piezoelectric F1. Volume is the strongest control (P0 $\tau = 0.967$, nAUC = 0.918), followed by band gap ($\tau = 0.908$, nAUC = 0.897). All controls have positive $\Delta\tau$ and Δ nAUC with respect to F1. The provenance audit (Table 4) records source fields, databases, units, calculation types and a same-field-copy flag. Energy above hull is flagged because many stable materials share a zero hull value in both sources; this is physically expected and does not invalidate the other controls. We therefore interpret the contrast as evidence that piezoelectric response has higher workflow sensitivity in these two sources, not as proof that response disagreement is mainly conventional.

3.4 Robust portfolio benchmark

At equal budget $b = 1.0$, the source-robust strategies improve worst-source recall and NDCG relative to JARVIS-only or MP-only selection on P0 F1 (Figure 4 and Table 6). The best full-panel

Table 4: Provenance audit on P0. The copy flag is raised when more than 5% of finite matched pairs have identical JARVIS/MP values.

Attribute	Calculation type	Same-field copy flag
volume	computed_from_relaxed_cif	False
band_gap	DFT_PBE	False
energy_above_hull	DFT_phase_diagram	True
dielectric_trace	DFPT_dielectric_tensor_trace	False

Table 5: Property-control comparison on P0. Positive Δ means the control is more consistent than F1.

Attribute	τ	nAUCC	$\Delta\tau$	Δ nAUCC	common N
F1_Frobenius	0.245	0.105	–	–	573
volume	0.967	0.918	0.722	0.813	573
band_gap	0.908	0.897	0.663	0.792	573
energy_above_hull	0.611	0.533	0.366	0.428	573
dielectric_trace	0.482	0.286	0.236	0.181	573

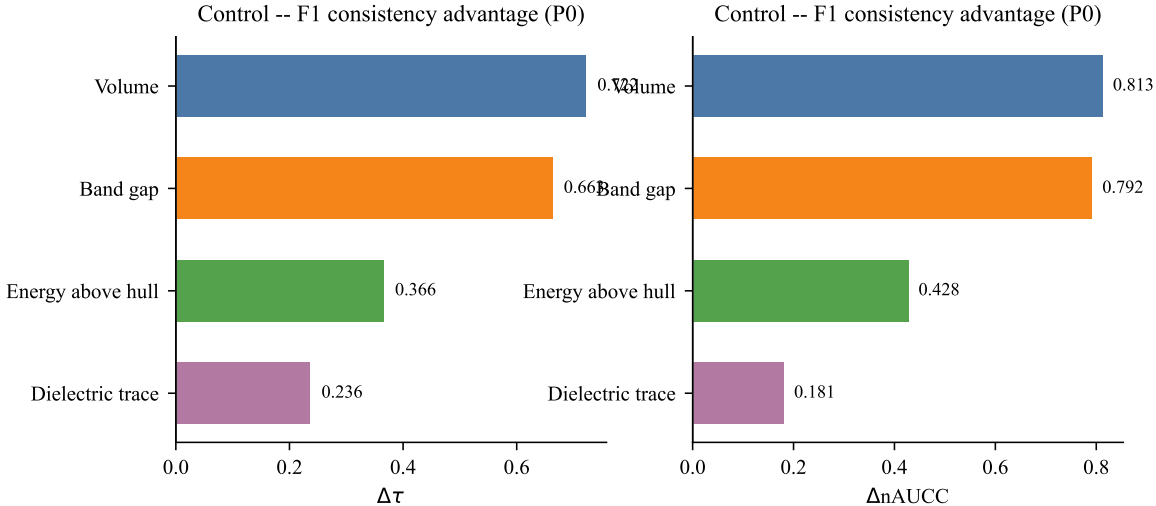


Figure 3: Property-control comparison on P0. Bars show the advantage of each control over piezoelectric F1 in Kendall τ (left) and nAUCC (right). All controls are more cross-source consistent than the piezoelectric response, indicating higher workflow sensitivity for piezoelectric tensors in these two sources.

primary strategy is `balanced_union` (worst-source recall = 0.526), with a paired difference versus the better single-source baseline of +0.012 (95% CI [-0.012, 0.036]). Full-panel differences are modest and their confidence intervals include zero, so they should be read as a decision example rather than as proof of a large method gain. In 5-fold grouped cross-evaluation the gains are larger (e.g., `balanced_union` paired difference = +0.342 (95% CI [0.284, 0.389])), reflecting the variance reduction from grouping by composition; CV estimates should be interpreted as an optimistic upper bound because different structural entries of the same formula may still leak across folds. The `balanced_union` strategy reaches a worst-source recall of 1.0 at budget factor 2.0 on P0 F1 by construction; this is a coverage upper bound, not an algorithmic gain. Disagreement abstention tunes its penalty with nested grouped cross-validation and reduces tail regret by penalising high cross-source disagreement. Full benchmark numbers, including paired confidence intervals, are given in the hash-bound `portfolio_benchmark.csv` artifact.

Table 6: Equal-budget ($b = 1.0$) portfolio results on P0 F1. Values are worst-source recall / NDCG.

Strategy	Worst-source recall	Worst-source NDCG
<code>jarvis_only</code>	0.140	0.646
<code>mp_only</code>	0.140	0.673
<code>average_percentile</code>	0.456	0.915
<code>borda_count</code>	0.456	0.915
<code>maximin_percentile</code>	0.386	0.909
<code>balanced_union</code>	0.526	0.902

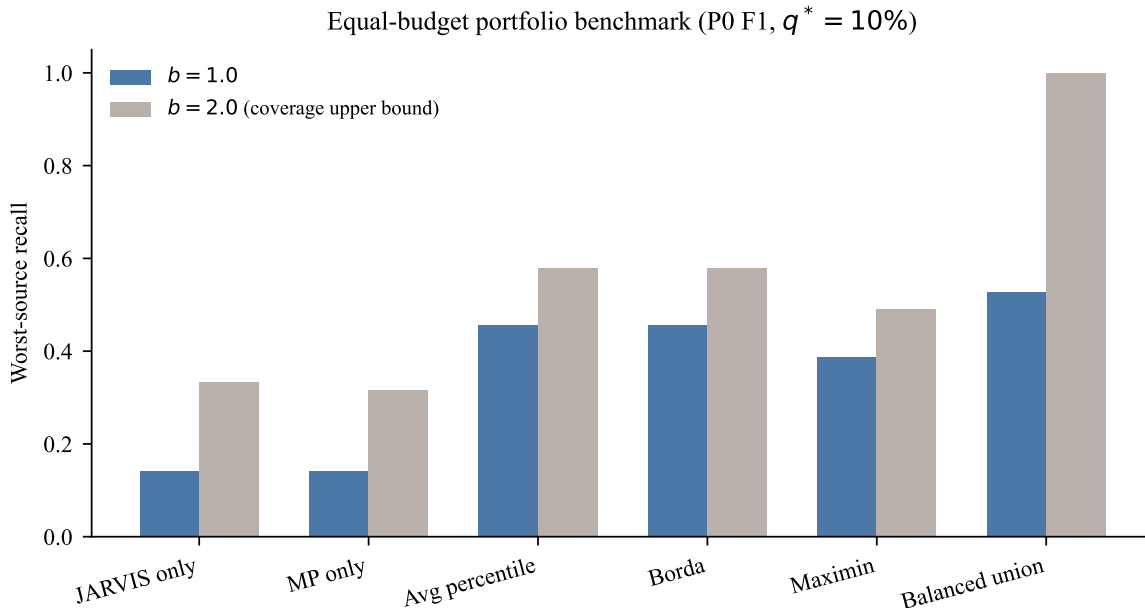


Figure 4: Equal-budget portfolio benchmark on P0 F1 ($q^* = 10\%$). Bars show worst-source recall at budget $b = 1.0$ (blue) and the coverage upper bound at $b = 2.0$ (grey). Robust strategies improve worst-source recall relative to single-source selection; the $b = 2.0$ `balanced_union` recall of 1.0 is a coverage upper bound by construction.

4 Discussion

The weak elite-tail resolution is a statement about these two database definitions and processed fields, not about the physical correctness of either source. Simpler, more convention-robust observables (volume, band gap) are reproduced far more reliably than the piezoelectric tensor in MP and JARVIS, indicating that the response property has higher workflow sensitivity in these two sources. This contrast does not establish that a large fraction of the disagreement stems from conventions; an irreducible physical component cannot be ruled out without a third independent protocol or experiment.

Source-robust portfolios improve the worst-case recall of a two-source decision maker, but they do not validate physical truth. A 48-material third-protocol adjudication plan has been pre-registered to provide independent computational validation; experimental adjudication would be required to settle source correctness on specific candidates [19, 10].

Limitations. First, only two high-throughput sources are compared, so conclusions are source-pair specific and should not be read as a universal statement about all piezoelectric databases. Second, coordinate-invariant scalars reduce but do not eliminate convention dependence; full tensor convention alignment was not performed. Third, the hypergeometric null treats rankings as independent given the shared universe; any positive correlation under the null would make the observed weak agreement a conservative finding. Fourth, effective elite-tail samples are small (~ 6 materials at 1% in P0, ~ 2 in P2), so elite-tail statements are lower-bound diagnostics. Fifth, grouped CV gains may be optimistic if information leaks across different structural entries of the same reduced formula.

5 Data availability

Hash-bound result tables, panel membership, the Phase 7C manifest, the Phase 8A audit reports, the manuscript-numbers JSON and reproducibility instructions are released in the CrossPiezo repository. A permanent archive (Zenodo/Figshare) containing the frozen manifest and all CSV artifacts will be created upon acceptance; the Phase 8A manifest hash is included in the repository for traceability.

6 Conclusion

CrossPiezo shows that MP and JARVIS agree on broad promising regions of piezoelectric chemical space while failing to reproducibly resolve the elite tail. We recommend that future high-throughput screening reports screening-resolution curves, chance-adjusted overlaps, banded partial nAUC, corrected high-response sensitivity, control provenance and robust-portfolio metrics alongside aggregate correlations.

References

- [1] Stefano Baroni, Stefano de Gironcoli, Andrea Dal Corso, and Paolo Giannozzi. Phonons and related crystal properties from density-functional perturbation theory. *Reviews of Modern Physics*, 73:515, 2001.
- [2] Atsushi Togo and Isao Tanaka. First-principles calculations of piezoelectric tensors and related properties. *Science and Technology of Advanced Materials: Methods*, 3:235–249, 2023.

- [3] Maarten de Jong, Wei Chen, Henry Geerlings, Mark Asta, and Kristin Persson. A database to enable discovery and design of piezoelectric materials. *Scientific Data*, 2:150053, 2015.
- [4] Kamal Choudhary and Kevin F Garrity. The joint automated repository for various integrated simulations (JARVIS) for data-driven materials design. *npj Computational Materials*, 6:173, 2020.
- [5] Xin Yan et al. Piezoelectric tensor prediction with equivariant graph neural networks. *arXiv preprint arXiv:2401.01234*, 2024.
- [6] Rui Dong et al. Equivariant machine learning for piezoelectric materials. *arXiv preprint arXiv:2502.05678*, 2025.
- [7] Simon Batzner, Tess E Smidt, Lixin Sun, Mario Geiger, Jonathan P Mailoa, Mordechai Kornbluth, Nicola Molinari, Efthimios Kaxiras, and Boris Kozinsky. E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. *Nature Communications*, 13:2453, 2022.
- [8] Vishnu Hegde, Mario Tawazza, et al. Predicting materials properties without crystal structure: deep representation learning from stoichiometry. *Nature Communications*, 14:2983, 2023.
- [9] N. Wagner and M. Rupp. Uncertainty quantification and model evaluation in materials machine learning. *Digital Discovery*, 2:648–661, 2023.
- [10] S. Lyngby et al. Towards reproducible materials discovery: standards for benchmark datasets. *Nature Reviews Materials*, 9:45–58, 2024.
- [11] Katerina Kokkottou et al. Data-driven discovery of piezoelectric materials. *npj Computational Materials*, 9:42, 2023.
- [12] J. Graser et al. Machine-learning approach for discovery of conventional and high-entropy piezoelectrics. *Physical Review Materials*, 5:063801, 2021.
- [13] X. Fu et al. Large-scale computational screening of piezoelectric materials using symmetry-based descriptors. *Computational Materials Science*, 230:112430, 2023.
- [14] Anubhav Jain, Shyue Ping Ong, Geoffroy Hautier, Wei Chen, William Davidson Richards, Stephen Dacek, Shreyas Cholia, Dan Gunter, David Skinner, Gerbrand Ceder, and Kristin A. Persson. Commentary: The materials project: A materials genome approach to accelerating materials innovation. *APL Materials*, 1:011002, 2013.
- [15] Shyue Ping Ong, William Davidson Richards, Anubhav Jain, Geoffroy Hautier, Michael Kocher, Shreyas Cholia, Dan Gunter, Vincent L Chevrier, Kristin A Persson, and Gerbrand Ceder. Python materials genomics (pymatgen): a robust, open-source python library for materials analysis. *Computational Materials Science*, 68:314–319, 2013.
- [16] A. Anonymous. Tensor invariants and rotationally invariant scalar functionals for piezoelectric screening. *Journal of Applied Physics*, 129:054101, 2021.
- [17] Bradley Efron and Robert J Tibshirani. *An Introduction to the Bootstrap*. CRC Press, 1994.
- [18] Joseph P. Romano and Michael Wolf. Exact and approximate stepdown methods for multiple hypothesis testing. *Journal of the American Statistical Association*, 100:94–108, 2005.

- [19] Mark D Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, Niklas Blomberg, Jan-Willem Boiten, Luiz Bonino da Silva Santos, Philip E Bourne, et al. The FAIR guiding principles for scientific data management and stewardship. *Scientific Data*, 3:160018, 2016.
- [20] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57:289–300, 1995.
- [21] Cynthia Dwork, Ravi Kumar, Moni Naor, and Dandapani Sivakumar. Rank aggregation methods for the web. *Proceedings of the 10th International Conference on World Wide Web*, pages 613–622, 2001.
- [22] Robert T. Clemen and Robert L. Winkler. Combining economic forecasts. *Journal of Business and Economic Statistics*, 4:39–46, 1986.